THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 10 — MATHS · EASY LEVEL FULL-YEAR PAPER SET 1

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. The HCF of 12 and 18 is: 1

(a) 12 **(b)** 6 **(c)** 36 **(d)** 3

2. If $p(x) = x^2 - 3x + 2$, then $p(4)$ is: 1

(a) 6 **(b)** 30 **(c)** 8 **(d)** 4

3. The pair of equations $x + 2y = 3$ and $2x + 4y = 6$ has: 1

(a) infinitely many solutions (coincident lines)

(b) exactly two solutions

(c) no solution (parallel lines)

(d) exactly one solution (intersecting lines)

4. Which of the following is a quadratic equation? 1

(a) $3x + 2 = 0$ **(b)** $x + 3 = 2$ **(c)** $x^2 + 3x + 2 = 0$ **(d)** $x^3 + 3x + 2 = 0$

5. The common difference of the AP $3, 7, 11, 15, \ldots$ is: 1

(a) 8 **(b)** 4 **(c)** -4 **(d)** 5

6. In a right triangle with legs 3 and 4, the hypotenuse is: 1

(a) 25 **(b)** 1 **(c)** 7 **(d)** 5

7. The distance of the point $(3, 4)$ from the origin is: 1

(a) 7 **(b)** 5 **(c)** 1 **(d)** 25

8. In a right triangle with perpendicular 4, base 3 and hypotenuse 5, $\sin\theta =$ 1

(a) $\frac{3}{5}$ **(b)** $\frac{5}{4}$ **(c)** $\frac{4}{5}$ **(d)** $\frac{4}{3}$

9. From a point on level ground, 7 m away from the foot of a vertical tower, the angle of elevation of its top is 45°. The height of the tower is: 1

(a) $7\sqrt{3}$ m **(b)** 7 m **(c)** $\frac{7\sqrt{3}}{3}$ m **(d)** 14 m

10. A point P is 5 cm from the centre O of a circle of radius 3 cm. The length of the tangent drawn from P to the circle is: 1

(a) 2 cm **(b)** 8 cm **(c)** 4 cm **(d)** 16 cm

11. The radius of a circle is 7 cm. The area of the circle is: 1

(a) 44 cm² **(b)** 308 cm² **(c)** 154 cm² **(d)** 22 cm²

12. A right circular cylinder has base radius 7 cm and height 5 cm. The curved surface area of the cylinder is: 1

(a) 770 cm² **(b)** 440 cm² **(c)** 220 cm² **(d)** 528 cm²

13. The class interval is 10–20. The class mark is: 1

(a) 15 **(b)** 20 **(c)** 10 **(d)** 16

14. A die is thrown once. The probability of getting an even number is: 1

(a) $\frac{1}{3}$ **(b)** $\frac{1}{2}$ **(c)** $\frac{2}{3}$ **(d)** $\frac{3}{7}$

15. The LCM of 4 and 6 is: 1

(a) 10 **(b)** 2 **(c)** 24 **(d)** 12

16. The zero of the linear polynomial $2x - 8$ is: 1

(a) -4 **(b)** -8 **(c)** 8 **(d)** 4

17. Which of the following pairs (x, y) is a solution of $2x + 3y = 12$? 1

(a) $(3, 2)$ **(b)** $(-3, 2)$ **(c)** $(3, -2)$ **(d)** $(2, 3)$

18. Which of the following is a root of $x^2 - 5x + 6 = 0$? 1

(a) 5 **(b)** -2 **(c)** 1 **(d)** 2

19. The 10th term of the AP $3, 7, 11, 15, \ldots$ is: 1

(a) 39 **(b)** 43 **(c)** 40 **(d)** 35

20. The ratio of corresponding sides of two similar triangles is 1:3. Their perimeters are in the ratio 1: 1

(a) 3 **(b)** 1 **(c)** 9 **(d)** 4

SECTION — B (Very Short Answer Type) 6 × 2 = 12

Answer in about 20–30 words each.

21. Find the HCF of 18 and 24 by prime factorisation. 2

22. Find the value of $2x^2 - 3x + 1$ at $x = 3$. 2

23. Check whether the pair $2x + 3y = 7$, $4x + 6y = 14$ is consistent. 2

24. Check whether $x = 2$ is a root of $x^2 - 7x + 10 = 0$. 2

25. Write the first four terms of the AP with $a = 5$ and $d = 3$. 2

26. Find the hypotenuse of a right triangle with legs 6 cm and 8 cm. 2

SECTION — C (Short Answer Type) 6 × 3 = 18

Answer in about 40–60 words each.

27. Find the LCM and HCF of 26 and 91 by prime factorisation. 3

28. Find the zeroes of $x^2 - 8x + 15$ by splitting the middle term and verify the sum of zeroes. 3

29. Solve by elimination: $x + y = 8$ and $x - y = 2$. 3

30. Solve by factorisation: $x^2 - 5x + 6 = 0$. 3

31. Find the 12th term of the AP $3, 7, 11, 15, \ldots$ 3

OR

In a right triangle the two perpendicular sides are 3 cm and 4 cm. Find the hypotenuse and the perimeter. 3

32. Find the distance between the points $(-2,\ 1)$ and $(1,\ 5)$. 3

OR

In $\triangle ABC$, right-angled at B, $AB = 3$ cm and $BC = 4$ cm. Find $\sin A$, $\cos A$ and $\tan A$. 3

SECTION — D (Long Answer Type) — 6 × 5 = 30

Answer in about 100–150 words each.

33. Find the HCF and LCM of 135 and 225 by prime factorisation and verify that HCF $\times$ LCM $=$ product. 5

34. Find the zeroes of $x^2 - 5x + 6$ and verify the relationship between the zeroes and the coefficients. 5

35. Solve by elimination: $2x + 3y = 12$ and $5x + 4y = 23$. 5

36. Solve by factorisation: $2x^2 - 7x + 3 = 0$. 5

OR

How many terms are there in the AP $7, 10, 13, \ldots, 64$? 5

37. A ladder 5 m long reaches a window 4 m above the ground. Find the distance of the foot of the ladder from the wall. 5

OR

Find the area of the triangle with vertices $(0,\ 0)$, $(4,\ 0)$ and $(0,\ 3)$. 5

38. If $\tan\theta = \dfrac{4}{3}$, find $\sin\theta$, $\cos\theta$ and $\sec\theta$. 5

OR

Two points on level ground lie on opposite sides of a tower of height $6\sqrt{3}$ m, in line with its foot. The angles of elevation of the top of the tower from the two points are 30° and 60°. Find the distance between the two points. 5

END OF QUESTION PAPER — Total: 80 Marks

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